

1. CORE INSPECTION METADATA

Timestamp	2026-09-23T22:02:49.232973
Primary Finding	DENT
Severity Layer	HIGH
Detections Logged	1
Analysis Engine	YOLOv11-Nano (Local Edge)

2. ACQUIRED IMAGERY & BOUNDING BOXES



Fig 1. AI-annotated structural imagery overlay.

3. RAW DETECTION COORDINATES

ID	DEFECT CLASS	CONFIDENCE	SEVERITY	BOUNDING BOX [X, Y, W, H]
01	DENT	86.0%	HIGH	[0.430, 0.418, 0.310, 0.207]

4. RAG CONTEXT RETRIEVAL

Query N/A

Source Manual ac_43.13-1b_w-chg1.pdf (Page 150)

Extracted Snippet: *"sustain these stresses, and then allow them to return into the structure. The repair must be equal to the original structure, but not stronger or stiffer, which will cause stress concentrations or alter the resonant frequency of the structure. c. All-metal aircraft are made of very thin sheet metal, and it is possible to restore the strength of the skin without restoring its rigidity. All repairs should be made using the same type and thickness of material that was..."*

5. SYNTHESIZED REPAIR PROTOCOL

Generated via Local Phi-3 Mini strictly grounded in retrieved maintenance documentation.

- 1 Isolate the damaged area**
Clearly mark and isolate the high-severity dent on the aircraft's surface to define the extent of the repair needed.
- 2 Remove the dent**
Carefully remove the dented metal using appropriate tools, ensuring not to cause additional damage to the surrounding area.
- 3 Prepare the surface**
Clean and prepare the area where the dent was removed, ensuring it is free from debris and ready for repair.
- 4 Apply the repair material**
Select the same type and thickness of metal used in the original construction and apply it to the prepared area, following the manufacturer's guidelines for curing or setting.
- 5 Finish and inspect the repair**
Once the repair material has set, finish the repair to match the surrounding area's texture and inspect the repair to ensure it meets the required strength and appearance standards.